

P109 T45 (1) 证: $P(A) \cap P(B) = P(A \cap B)$

取 $x \in P(A) \cap P(B)$ 则有 $x \in P(A) \wedge x \in P(B)$

故有 $x \subseteq A \wedge x \subseteq B$

故 $x \subseteq A \cap B$ 故 $x \in P(A \cap B)$ 得 $P(A) \cap P(B) \subseteq P(A \cap B)$ ①

反之. 取 $x \in P(A \cap B)$ 有 $x \subseteq A \cap B$

故有 $x \subseteq A \wedge x \subseteq B$

故有 $x \in P(A) \wedge x \in P(B)$

也即 $x \in P(A) \cap P(B)$ 得 $P(A \cap B) \subseteq P(A) \cap P(B)$ ②

综合 ①② 得 $P(A \cap B) = P(A) \cap P(B)$

(2) 证 $P(A) \cup P(B) \subseteq P(A \cup B)$

取 $x \in P(A) \cup P(B)$ 则有 $x \in P(A) \vee x \in P(B)$

故有 $x \subseteq A \vee x \subseteq B$

故有 $x \subseteq (A \cup B)$

故有 $x \in P(A \cup B)$

非充要

T46 (1) $A \cup B = A \Leftrightarrow B \subseteq A$

(2) $A - B = A \Leftrightarrow A \cap B = \emptyset$

(3) $A - B = B \Leftrightarrow A = B = \emptyset$

$\downarrow$
 $\emptyset = (A \cap \sim B) \cap B = B \cap B$ 得 $B = \emptyset$. 代入得 $A = \emptyset$

(4) $A - B = B - A \Leftrightarrow A = B$

左推右: $(A - B) \cup A = (B - A) \cup A$ 得 $A = A \cup B$ 得 $A \subseteq B$

$(A \cap \sim B) \cup A = (B \cap \sim A) \cup A$

$A \cap (A \cup \sim B) = (B \cup A) \subseteq A$
又 $A \subseteq B \cup A$

同理 $B \subseteq A$

故 $A = B$

(5) $A \oplus B = A \Leftrightarrow B = \emptyset$

右推左显然. 左推右:

$A \oplus (A \oplus B) = A \oplus A = \emptyset$

$\therefore$ 由结合律得 $\emptyset \oplus B = \emptyset$ 得 $B = \emptyset$

(6) $A \oplus B = \emptyset \Leftrightarrow A = B$

$A \oplus (A \oplus B) = A \oplus \emptyset = A$

左 $= \emptyset \oplus B = B$. 右 $= A$

故 $B = A$.

或直接由定义 $A \oplus B = \emptyset$

得 $(A - B) \cup (B - A) = \emptyset$

故 $A - B = \emptyset$ 或 $B - A = \emptyset$

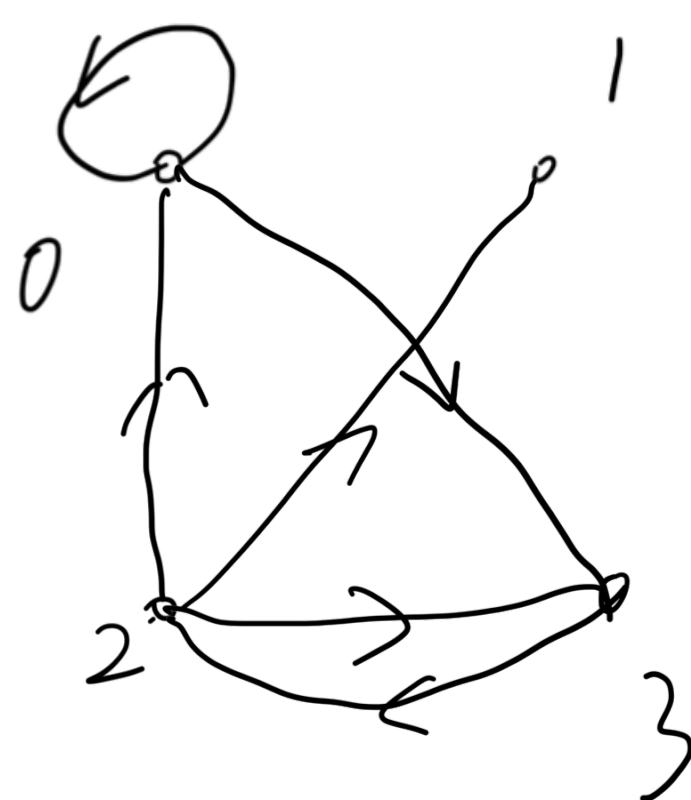
故 $A \subseteq B$ 或 $B \subseteq A$

Pr40 T12

R的关系矩阵:

$$\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

关系图:



T16

$A = \{a, b, c, d\}$

$R_1 = \{ \langle a, a \rangle, \langle a, b \rangle, \langle b, d \rangle \}$

$R_2 = \{ \langle a, d \rangle, \langle b, c \rangle, \langle b, d \rangle, \langle c, b \rangle \}$

$R_1 \circ R_2 = \{ \langle a, d \rangle, \langle a, c \rangle \}$

$R_2 \circ R_1 = \{ \langle c, d \rangle \}$

$R_1^2 = \{ \langle a, a \rangle, \langle a, b \rangle, \langle a, d \rangle \}$

$R_2^3 = \{ \langle b, c \rangle, \langle c, b \rangle, \langle b, d \rangle \}$